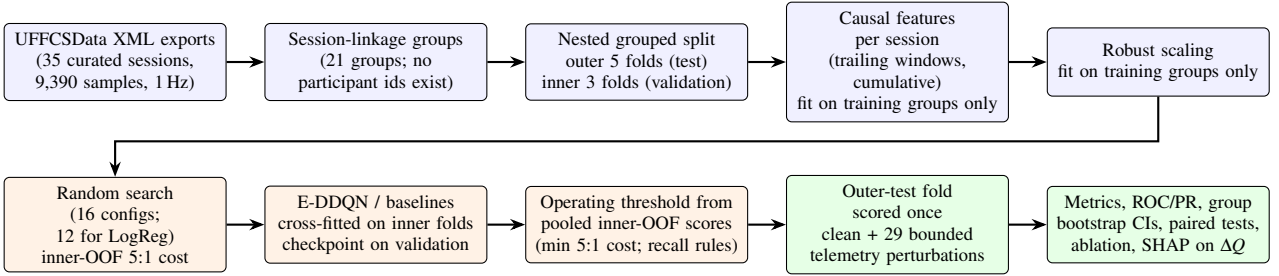
UFFCSData XML exports
(35 curated sessions,
9,390 samples, 1 Hz)
Session-linkage groups
(21 groups; no
participant ids exist)
Nested grouped split
outer 5 folds (test)
inner 3 folds (validation)
Causal features
per session
(trailing windows,
cumulative)
fit on training groups only
Robust scaling
fit on training groups only
Random search
(16 configs;
12 for LogReg)
inner-OOF 5:1 cost
E-DDQN / baselines
cross-fitted on inner folds
checkpoint on validation
Operating threshold from
pooled inner-OOF scores
(min 5:1 cost; recall rules)
Outer-test fold
scored once
clean + 29 bounded
telemetry perturbations
Metrics, ROC/PR, group
bootstrap CIs, paired tests,
ablation, SHAP on ΔQ